Dbms assessment

Bank table

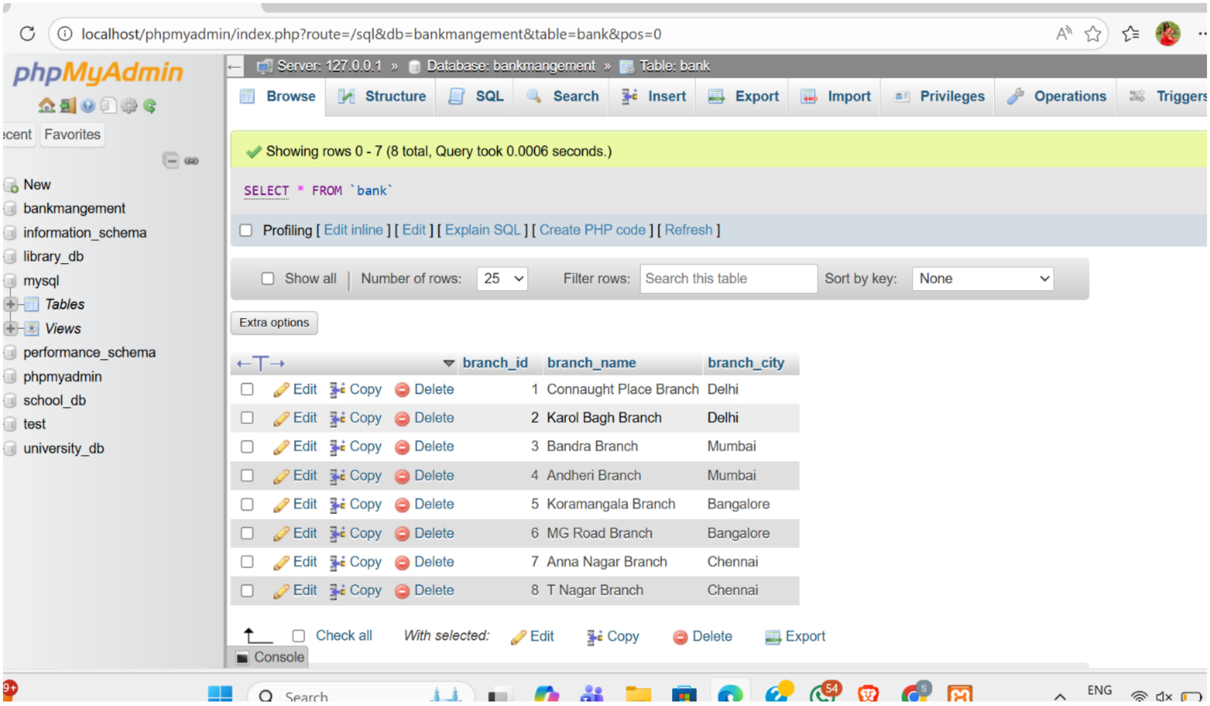
CREATE TABLE bank (

branch_id SERIAL PRIMARY KEY,

branch_name VARCHAR(100) NOT NULL,

branch_city VARCHAR(100) NOT NULL

);



Account Holder Table

CREATE TABLE account_holder (

account_holder_id SERIAL PRIMARY KEY,

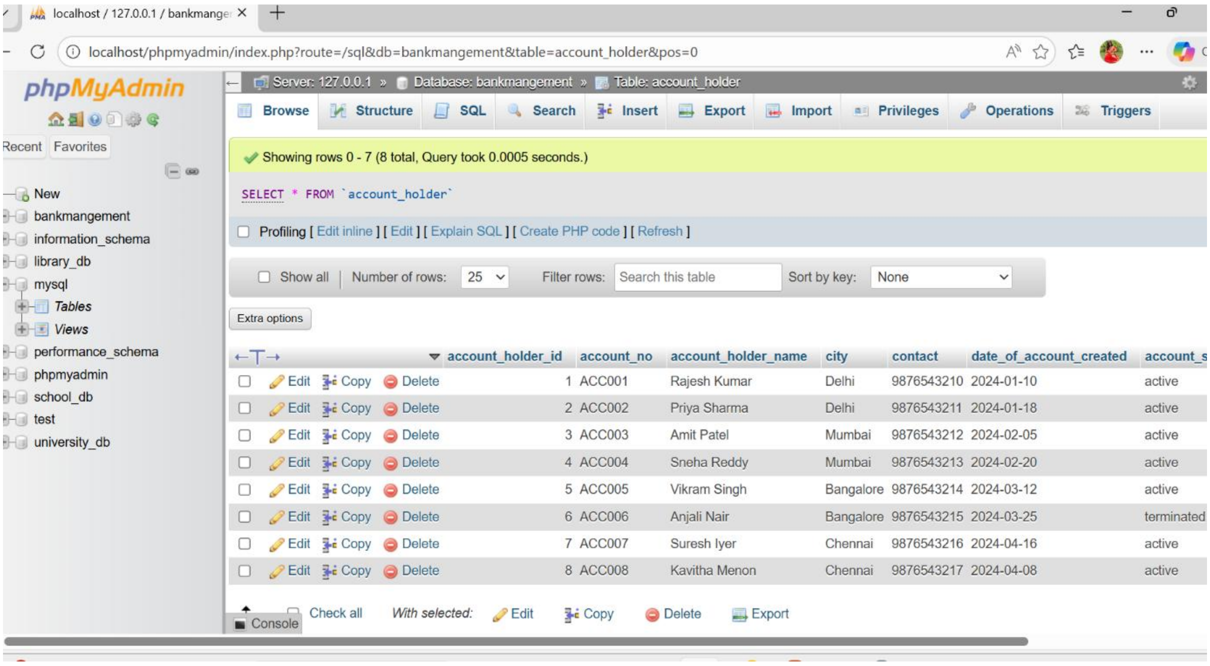
account_no VARCHAR(20) UNIQUE NOT NULL,

account_holder_name VARCHAR(100) NOT NULL,

city VARCHAR(100) NOT NULL,

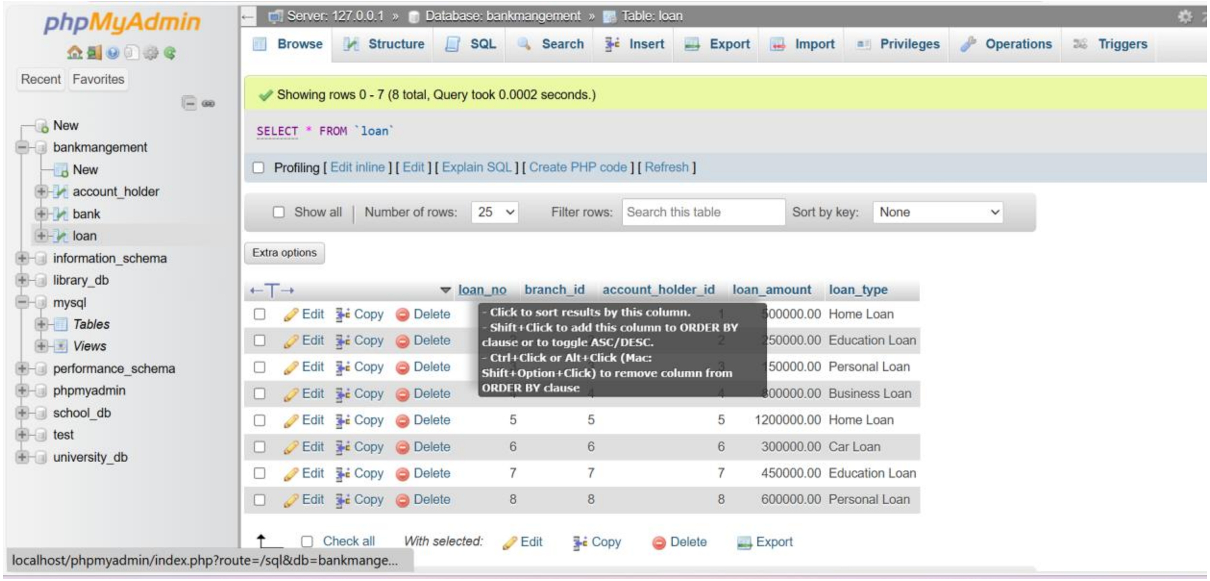
contact VARCHAR(15),

```
    date_of_account_created DATE DEFAULT CURRENT_DATE,
    account_status VARCHAR(20) CHECK (account_status IN ('active', 'terminated')),
    account_type VARCHAR(50),
    balance DECIMAL(15, 2) DEFAULT 0.00
);
```



Loan Table

```
CREATE TABLE loan (
    loan_no SERIAL PRIMARY KEY,
    branch_id INT REFERENCES bank(branch_id),
    account_holder_id INT REFERENCES account_holder(account_holder_id),
    loan_amount DECIMAL(15, 2) NOT NULL,
    loan_type VARCHAR(50)
);
```


phpMyAdmin
Recent Favorites
New
bankmangement
New
account_holder
bank
loan
information_schema
library_db
mysql
Tables
Views
performance_schema
phpmyadmin
school_db
test
university_db
Server: 127.0.0.1 » Database: bankmangement » Table: loan
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 7 (8 total, Query took 0.0002 seconds.)
SELECT * FROM `loan`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
loan_no branch_id account_holder_id loan_amount loan_type
- Click to sort results by this column.
- Shift+Click to add this column to ORDER BY clause or to toggle ASC/DESC.
- Ctrl+Click or Alt+Click (Mac: Shift+Option+Click) to remove column from ORDER BY clause
Edit Copy Delete Home Loan
Edit Copy Delete Education Loan
Edit Copy Delete Personal Loan
Edit Copy Delete Business Loan
Edit Copy Delete 5 5 5 1200000.00 Home Loan
Edit Copy Delete 6 6 6 300000.00 Car Loan
Edit Copy Delete 7 7 7 450000.00 Education Loan
Edit Copy Delete 8 8 8 600000.00 Personal Loan
Check all With selected: Edit Copy Delete Export
localhost/phpmyadmin/index.php?route=/sql&db=bankmange...

INTRA-BANK TRANSFER TRANSACTION

BEGIN;

UPDATE account_holder

SET balance = balance - 100

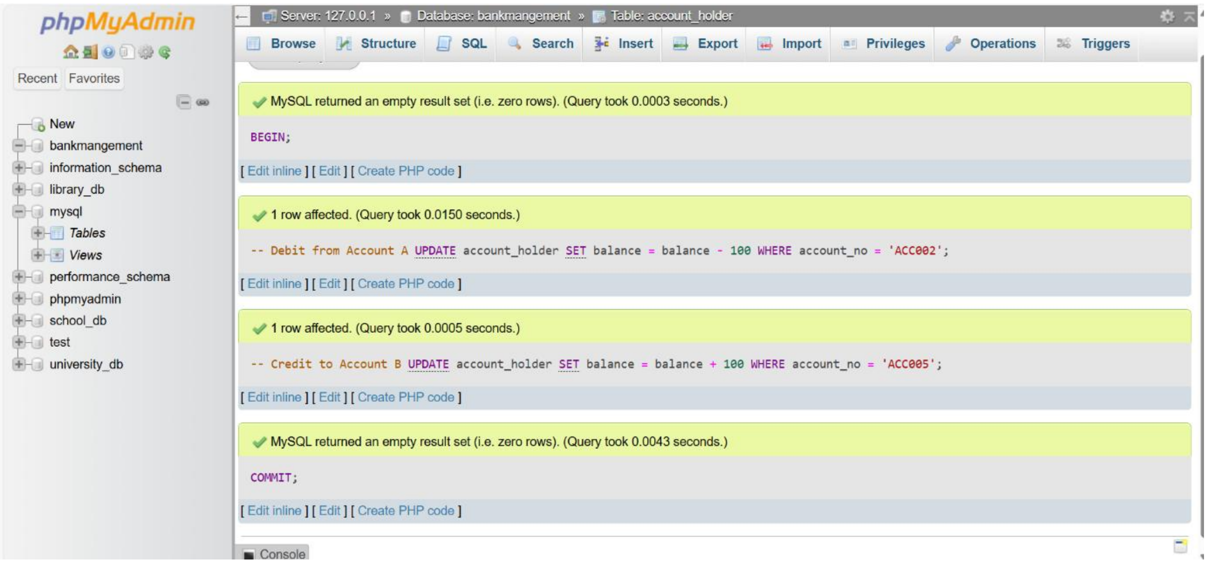
WHERE account_no = 'A';

UPDATE account_holder

SET balance = balance + 100

WHERE account_no = 'B';

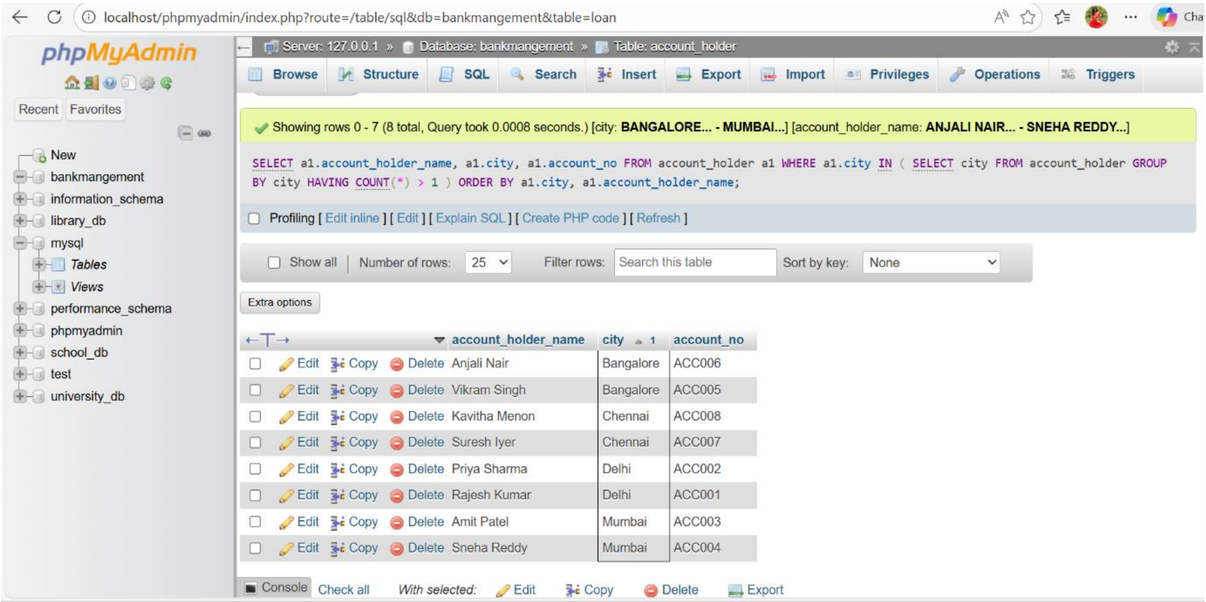
COMMIT;



Queries

1. Fetch account holders from the same city

```
SELECT a1.account_holder_name, a1.city, a1.account_no
FROM account_holder a1
WHERE a1.city IN (
    SELECT city
    FROM account_holder
    GROUP BY city
    HAVING COUNT(*) > 1
)
ORDER BY a1.city, a1.account_holder_name;
```



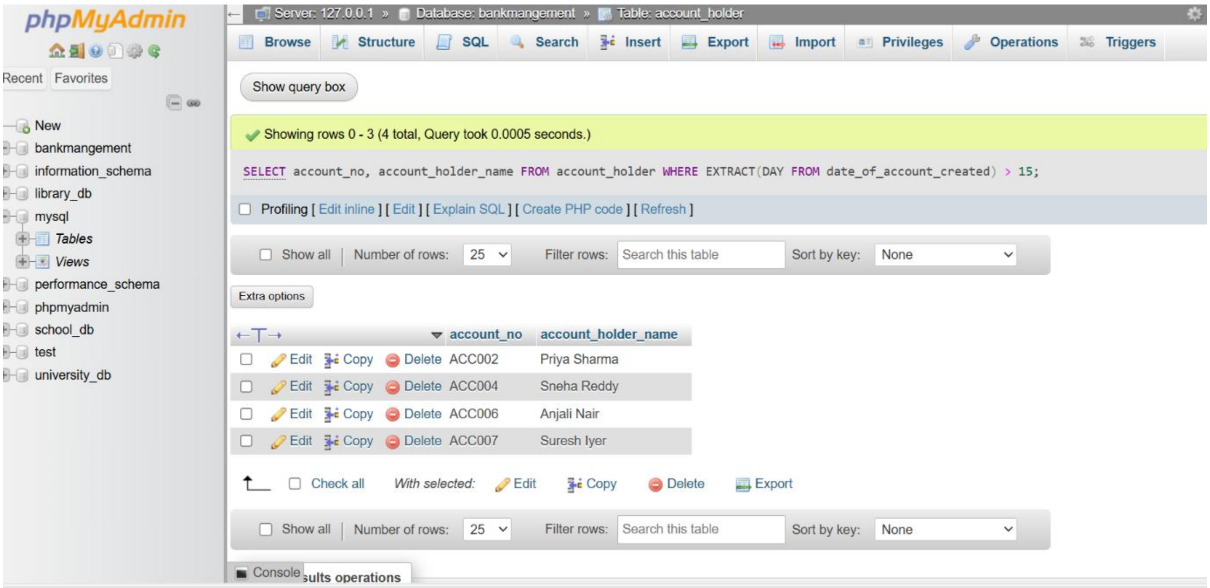
account_holder_name	city	account_no
Anjali Nair	Bangalore	ACC006
Vikram Singh	Bangalore	ACC005
Kavitha Menon	Chennai	ACC008
Suresh Iyer	Chennai	ACC007
Priya Sharma	Delhi	ACC002
Rajesh Kumar	Delhi	ACC001
Amit Patel	Mumbai	ACC003
Sneha Reddy	Mumbai	ACC004

2. Accounts created after 15th of any month

SELECT account_no, account_holder_name

FROM account_holder

WHERE EXTRACT(DAY FROM date_of_account_created) > 15;

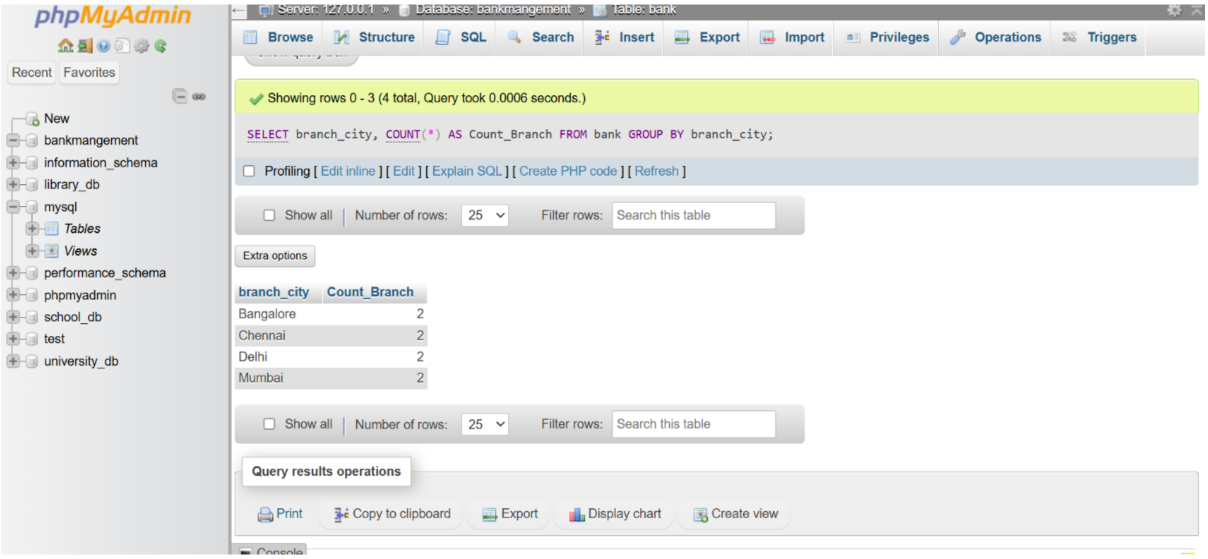


3. City name with branch count (alias: Count_Branch)

SELECT branch_city, COUNT(*) AS Count_Branch

FROM bank

GROUP BY branch_city;


phpMyAdmin
Recent Favorites
New
bankmangement
information_schema
library_db
mysql
Tables
Views
performance_schema
phpmyadmin
school_db
test
university_db
Server: 127.0.0.1 » Database: bankmangement » Table: bank
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)
SELECT branch_city, COUNT(*) AS Count_Branch FROM bank GROUP BY branch_city;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table
Extra options
branch_city Count_Branch
Bangalore 2
Chennai 2
Delhi 2
Mumbai 2
Show all | Number of rows: 25 Filter rows: Search this table
Query results operations
Print Copy to clipboard Export Display chart Create view
Console

4. Account holders with loans (using JOIN)

```
SELECT
    ah.account_holder_id,
    ah.account_holder_name,
    l.branch_id,
    l.loan_amount
FROM account_holder ah
JOIN loan l
    ON ah.account_holder_id = l.account_holder_id;
```